Incident Response Playbook

SSH Brute Force Attack — NIST SP 800-61 Rev. 2

Standard: NIST SP 800-61 Rev. 2 — Computer Security Incident Handling Guide
Incident Type: Credential Access — Brute Force (MITRE ATT&CK; T1110)
Severity: Medium
Status: Resolved
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Date: 2026-10-23

Incident Summary

Field	Value
Incident ID	IR-2026-001
Detection Time	2026-10-23 16:19:56 UTC
Containment Time	2026-10-23 16:20:14 UTC
Total Response Time	~18 seconds (automated)
Attack Type	SSH Brute Force
MITRE Technique	T1110 — Brute Force / Credential Access
Attacker IP	192.168.68.131 (Kali Linux VM)
Target IP	192.168.68.130 (Ubuntu Server — MrRobot)
Target Account	mrrobot
Detection Rule	Wazuh Rule 5758 — Maximum authentication attempts exceeded
Alert Level	8
Outcome	Detected and alerted. No successful authentication.

Phase 1 — Preparation

Security Controls in Place

Control	Tool	Configuration
Log collection	Wazuh Agent	/var/log/auth.log bind-mounted read-only into Wazuh Docker container
SIEM alerting	Wazuh 4.9.0	Rule 5758 enabled; alert level threshold set to 10 for email dispatch
Automated response	Shuffle SOAR	Webhook trigger → VirusTotal enrichment → email notification
IP reputation	VirusTotal v3	API integrated into Shuffle workflow for attacker IP lookup
Notification	Gmail SMTP	Automated email to SOC inbox on confirmed alert

Known Environment

```
[Kali Linux 192.168.68.131]        <- Attacker VM
[Ubuntu Server 192.168.68.130]     <- Target / Wazuh-monitored endpoint
[Wazuh Manager - Docker]           <- SIEM running on Ubuntu Server
[Shuffle SOAR - Docker]            <- Automation running on Ubuntu Server
```

Phase 2 — Detection & Analysis

2.1 Initial Detection

At **16:19:56 UTC**, Wazuh fired **Rule 5758** against the Ubuntu Server agent (MrRobot):

```
Rule ID:      5758
Description:  Maximum authentication attempts exceeded
Level:        8
Agent:        MrRobot (192.168.68.130)
Source IP:    192.168.68.131
Target User:  mrrobot
Log Source:   /var/log/auth.log
MITRE ID:     T1110
Tactic:       Credential Access
Technique:    Brute Force
```

2.2 Attack Reconstruction

The attacker (192.168.68.131) ran Hydra against the SSH service on the target:

```
hydra -l mrrobot -P /usr/share/wordlists/rockyou.txt ssh://192.168.68.130 -t 4 -V
```

• Wordlist: rockyou.txt (~14 million passwords)
• Threads: 4 concurrent SSH connections
• Target account: mrrobot
• Pattern: Rapid sequential failed authentication — matched Rule 5758 signature

2.3 Automated Enrichment (Shuffle SOAR)

Shuffle received the Wazuh webhook at 16:19:56 and completed enrichment by 16:20:14 (18 seconds total):

• **Node 1 — Webhook:** Received full Wazuh alert JSON payload
• **Node 2 — Change Me (Parser):** Extracted severity, rule ID, source IP, timestamp
• **Node 3 — VirusTotal v3:** Queried attacker IP 192.168.68.131

```
{
  "malicious": 0,
  "suspicious": 0,
  "undetected": 36,
  "harmless": 56,
  "timeout": 0
}
```

• **Node 4 — Email:** success: true — notification dispatched to SOC inbox

2.4 Threat Assessment

Factor	Assessment
Attacker IP reputation	Clean (VirusTotal: 0 malicious, 56 harmless)
Authentication success	No successful logins detected in auth.log
Lateral movement	No — attack contained to SSH on single host
Data exfiltration	No evidence
Persistence	No — no new accounts, cron jobs, or SSH keys observed
Severity	Medium — active brute force, no breach

Phase 3 — Containment, Eradication & Recovery

3.1 Immediate Containment

Action	Method	Status
Block attacker IP at firewall	ufw deny from 192.168.68.131	Recommended
Kill active SSH sessions	ss -tp | grep 192.168.68.131 → kill <pid>	Recommended
Disable SSH password auth	PasswordAuthentication no in sshd_config	Recommended

```
# Block attacker IP
sudo ufw deny from 192.168.68.131 to any

# Verify no active sessions
ss -tp | grep 192.168.68.131

# Restart SSH after config change
sudo systemctl restart sshd
```

3.2 Eradication — Verify No Breach

```
# Check for successful logins from attacker IP
grep "Accepted" /var/log/auth.log | grep "192.168.68.131"

# Check for new user accounts
grep "useradd\|adduser" /var/log/auth.log

# Check for new SSH authorized keys
cat /home/mrrobot/.ssh/authorized_keys
```

Expected result: No output — confirms no breach occurred.

3.3 Hardening Recommendations

Recommendation	Priority	Implementation
Disable SSH password auth; use key-based auth	High	PasswordAuthentication no in sshd_config

Recommendation	Priority	Implementation
Install and configure fail2ban	High	Auto-ban IPs after 5 failed attempts
Change SSH from default port 22	Medium	Port 2222 in sshd_config
Restrict SSH by IP allowlist	Medium	ufw allow from <trusted_ip> to any port 22
Enable MFA for SSH	Medium	libpam-google-authenticator

Phase 4 — Post-Incident Activity

4.1 Timeline of Events

Time (UTC)	Event
16:19:56	Hydra brute force begins from 192.168.68.131
16:19:56	Wazuh Rule 5758 fires — alert level 8
16:19:56	Shuffle webhook triggered
16:20:10	VirusTotal query returns — 0 malicious
16:20:14	Email alert delivered to SOC inbox
16:20:14	Incident detected, analysed, notified — total automated response: 18 seconds

4.2 Lessons Learned

What worked well:

• Wazuh Rule 5758 fired immediately — no manual review needed
• MITRE T1110 mapping was automatic — no analyst effort required for classification
• Shuffle pipeline completed enrichment and notification in 18 seconds
• VirusTotal enrichment gave instant IP reputation context

What needs improvement:

Gap	Impact	Fix
No automated blocking — alert only	Attacker could continue indefinitely	Add Shuffle node to run ufw deny via SSH on trigger
Alert threshold too low initially (level 3)	203+ emails from one scan	Set threshold before running attacks
No dedicated agent on attacker VM	Attacker-side visibility missing	Install agents on all VMs from the start
Email body is unformatted plain text	Hard to parse quickly	Improve Shuffle email template with structured HTML

4.3 Metrics

Metric	Value
Mean Time to Detect (MTTD)	< 1 second (automated rule)
Mean Time to Respond (MTTR)	18 seconds (automated enrichment + notification)

Metric	Value
False Positive Rate	0% — attack correctly classified
Alerts generated	1 (after threshold tuned to level 10)

Appendix — Evidence

Artifact	Description	Location
wazuh alert.png	Wazuh dashboard — Rule 5758 fired, level 8, MITRE T1110	GitHub repo
debug1.png	Shuffle execution overview — FINISHED, 18s runtime	GitHub repo
debug2.png	Change Me node — parsed Wazuh alert JSON payload	GitHub repo
debug3.png	VirusTotal result (status 200) + Email result (success: true)	GitHub repo
Email Alert.png	Gmail inbox — alert received with enrichment data	GitHub repo
Wazuh Enrichment.json	Full Shuffle workflow export — importable into any Shuffle instance	GitHub repo

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